CONTROL UNIT

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CONTROL UNIT(CU)

- CPU is partitioned into **ALU** and **CU.**
- It **controls the flow of data** between the processor, memory & peripherals (I/O device).
- The **Control Unit** (along with IR) **decodes/interpret** the machine language instruction and issues the **control signals** to make the CPU execute that instruction.
- **Function of Control Unit** is to generate relevant **timing & control signals** to all **operations** in the computer.

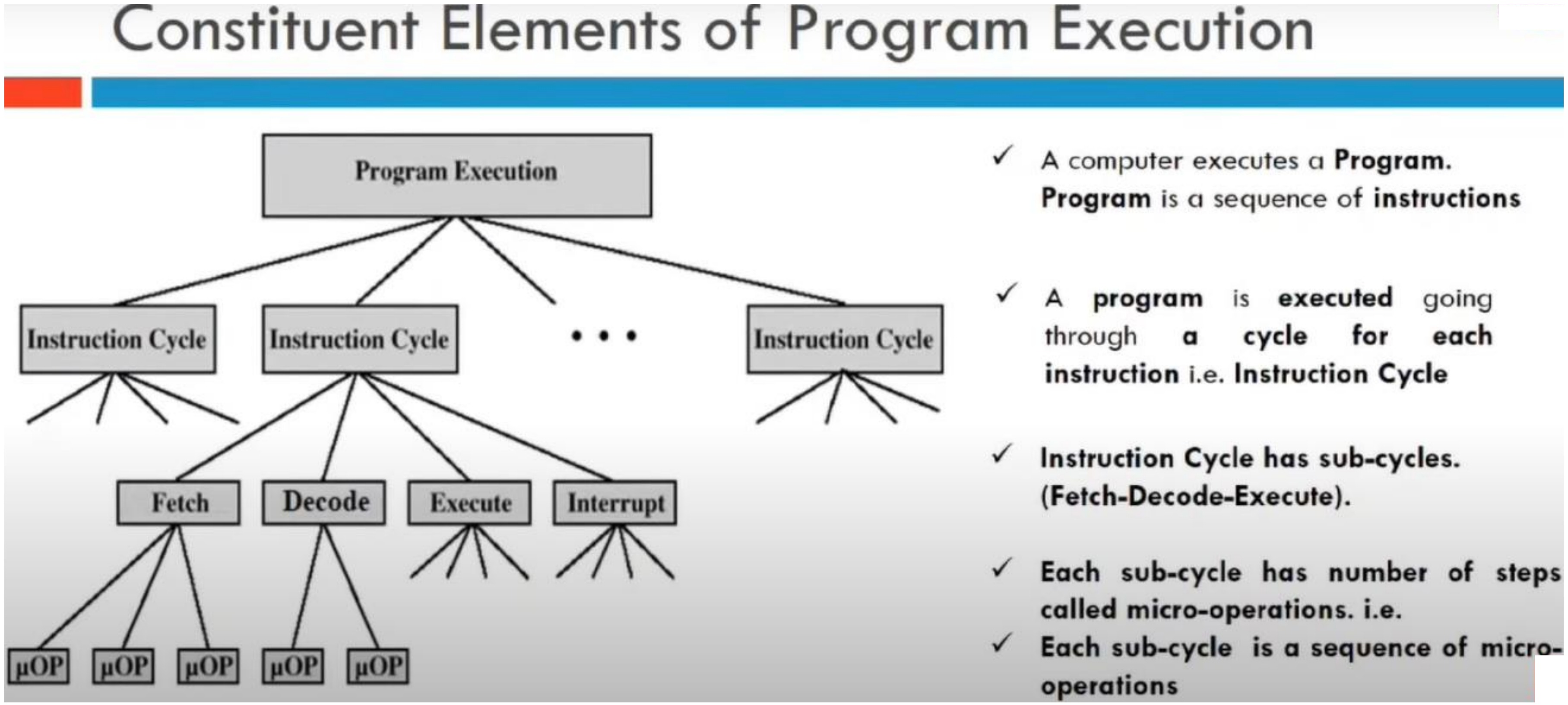
FUNCTIONS OF CONTROL UNIT

Two Basic Functions of Control Unit:

- <u>Sequencing</u>
  - CU causes the CPU to step through a series of micro-operations in proper sequence based on the **program** being executed
- <u>Execution</u>
  - CU causes each micro-operation to be executed.

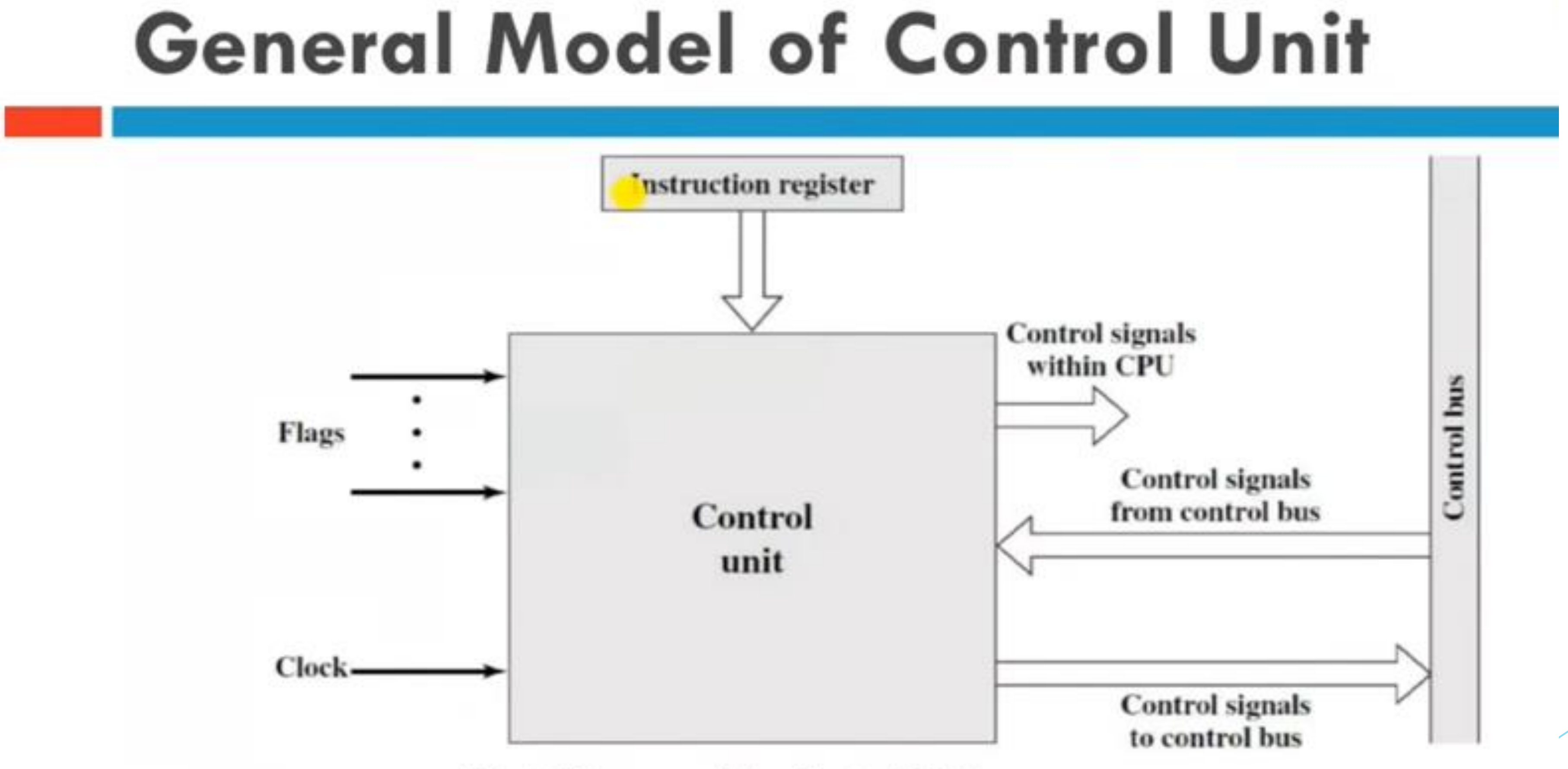
✓ This is done using <u>Control Signals</u>.

ELEMENTS OF PROGRAM EXECUTION



Microoperations are the atomic operations of the processor.

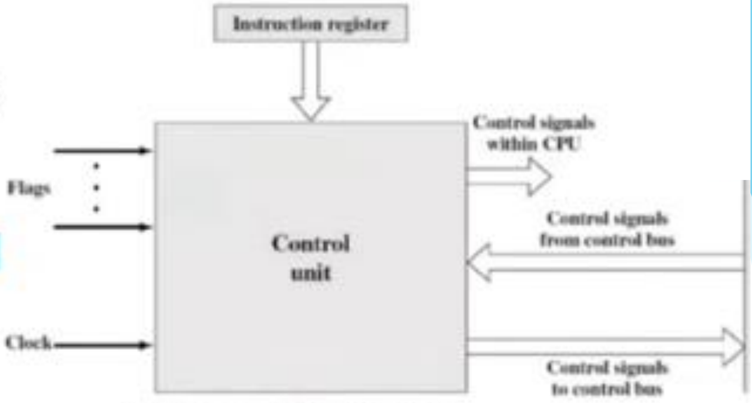
GENERAL MODEL OF CONTROL UNIT



Block Diagram of the Control Unit

Types of Control Signals

(Inputs and Outputs of CU)



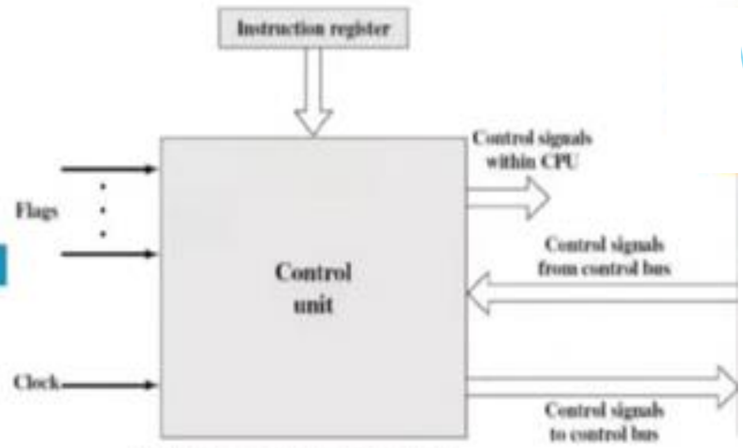
Block Diagram of the Control Unit

Control Unit Inputs:

- **Clock**
  - This is how the CU "keeps time".
  - CU causes one micro-operation (or set of parallel micro-operations) to be performed per clock pulse.
  - This is sometimes referred to as the processor cycle time, or the clock cycle time.
- **Instruction register**
  - The Opcode & addressing mode of the current instruction in IR, determines which micro-operations are performed.
- **Flags**
  - These are needed by the control unit
    - to determine the status of the processor, and
    - the outcome of previous ALU operations.
  - Example: For the *Increment-and-skip-if zero (ISZ) instruction*, the CU will increment the PC if the zero flag is set.
- **Control Signals from Control Bus**
  - Control bus portion of system bus provides signals to CU
  - Example:
    - Interrupts
    - Acknowledgements

Types of Control Signals

(Inputs and Outputs of CU)



Block Diagram of the Control Unit

Control Unit Outputs:

- **Control Signals within CPU**
  - Cause data movement from one register to another register
  - Activate specific ALU functions (add, sub, mul, etc)
- **Control Signals to the control bus**
  - Control signals to memory (via the system bus)
  - Control signals to I/O modules

IMPLEMENTATION OF CONTROL UNIT

Two Major types of Control Organization / Implementation of Control Unit

To execute an instruction, the Control Unit of the CPU must generate the required control signals in proper sequence.

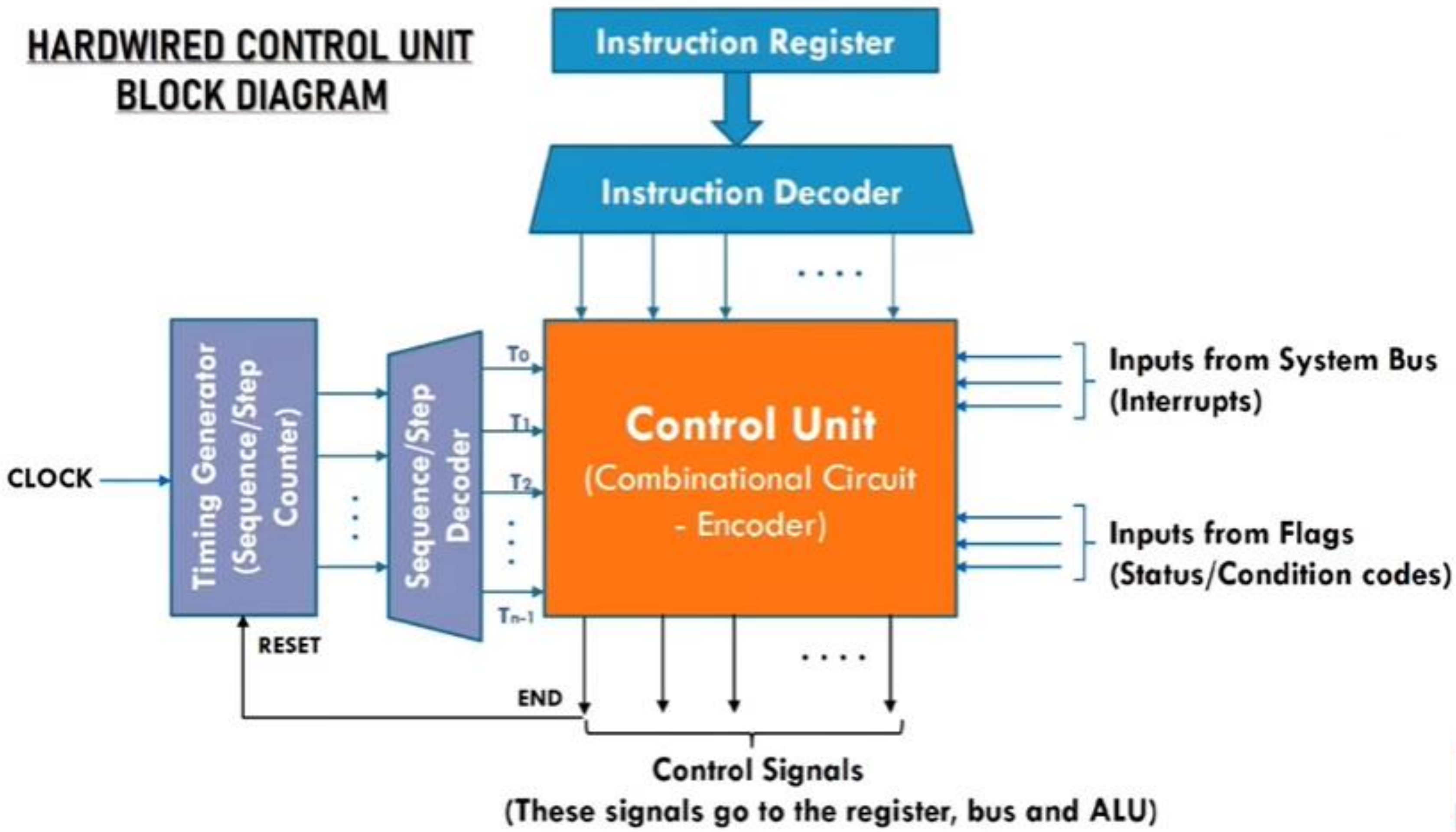
There are 2 approaches used for generating control signals in proper sequence:

1. **Hardwired Control Unit**
2. **Microprogrammed Control unit**

HARDWIRED CONTROL UNIT

- In a hardwired control unit, the control logic is implemented with logic gates, flip-flops, decoders, encoders and other digital logic circuits i.e. Control unit is implemented with Hardware.

BLOCK DIAGRAM OF HARDWIRED CONTROL UNIT



WORKING OF HARDWIRED CONTROL UNIT

Working

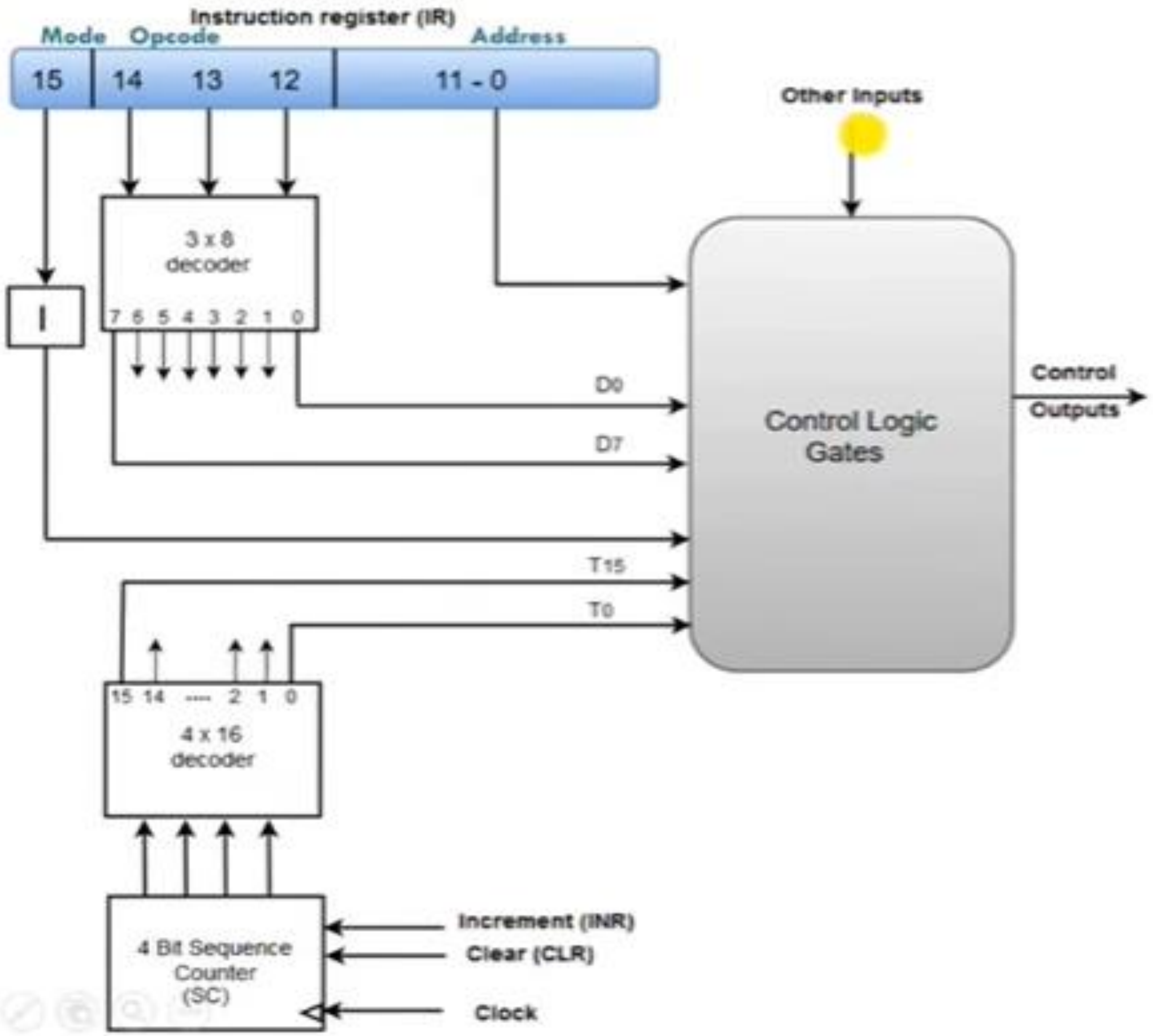
- **Hardwired control unit gets set of inputs (from IR, flags, clock, system bus)** and **transform them into a set of control signals.**
- **Instruction decoder** decodes the instruction loaded in **IR (Instruction Register).**
- **Sequence counter** provides separate signal line for each step or time slot, in a control sequence (T1, T2, T3,…..).
- After execution of each instruction "**END**" signal is generated this "**RESET**" the **control sequence counter** and make it ready for generation of control sequence for next instruction.
- Hardwired control units are generally **faster** and **simpler** in design, but they are also **less flexible** and **more difficult to modify or update.**
- **RISC (Reduced Instruction Set Computer)** are implemented with **Hardwired Control Unit.**

ADVANTAGE AND DISADVANTAGE OF HARDWIRED CONTROL UNIT

Advantage & Disadvantage

- **Advantage**
  - **Very fast** (Because signals are generated by combinational circuits).
    - RISC are implemented with Hardwired Control Unit.
- **Disadvantage:**
  - Instruction set and control logic are tied together by special circuits that are **complex and difficult to design or modify.**
    - If someone designs a hardwired computer and later decides to extend instruction set, the physical components in the computer must be **changed.**
    - This is prohibitively expensive, because not only new chips must be fabricated but also old one must be located and replaced.

Hardwired Control Unit of Basic Computer



- Two decoders, sequence counter and logic gates make up a **Hardwired Control.**

- The **instruction register (IR)** stores an instruction retrieved from the memory unit.
- An **instruction register** (IR) consists of ***the operation code (12-14), the I bit (15), and bits 0 through 11.***
- The **operation code bits (12-14)** are decoded with 3 x 8 decoder.
- The **8 decoder's outputs** (**D0 through D7**) goes to **control logic gates** to perform specific operation.
- The last **bit 15** is transferred to a **flip-flop** designated by symbol I.
- The **first 12 bits (0 to 11)** are applied to **control logic gates.**
- The **4-bit sequence counter (SC)** can count from 0 to 15 in binary.
- Its output is decoded into 16 timing pulses from **T_0 through T_{15} with 4x 16 decoder.**
- The **SC** is incremented by **INR** input or cleared by **CLR** input synchronously,.

MICROPROGRAMMED CONTROL UNIT

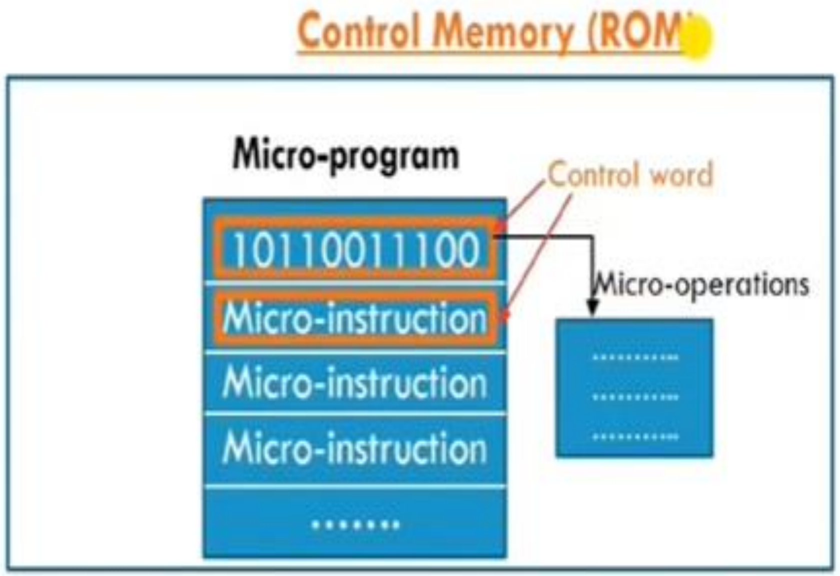
Microprogrammed Control unit

- A **Microprogrammed Control unit** is implemented using **programming approach**.
- In **Microprogrammed Control unit,** the control information (control signals) is stored in a **Control Memory**. Control memory is programmed to initiate the required sequence of micro-operations.
  - **Control Memory** contains **microprogram** which is sequence of **microinstructions** that specify sequence of **micro-operations**.

Microprogrammed Control unit

- A <u>**Microprogrammed Control unit**</u> is a control unit in which,
  - The **control signals** (control information) are stored in **binary** form (1's & 0's) as **Control Word** in a ROM chip called **Control Memory**.
  - Each control word in control memory contains within it a **microinstruction.**
  - The microinstruction specifies one or more **micro-operations** for the system.
  - A sequence of microinstructions is called a **micro-program.**
  - **Execution of microinstruction** is responsible for generation of a set of **control signals.**
  - The control memory is assumed to be in **Read Only Memory (ROM)** within which all control information is permanently stored.
  - The address where these microinstruction are stored in control memory is generated by **microprogram sequencer** with the help of IR (Instruction Register).

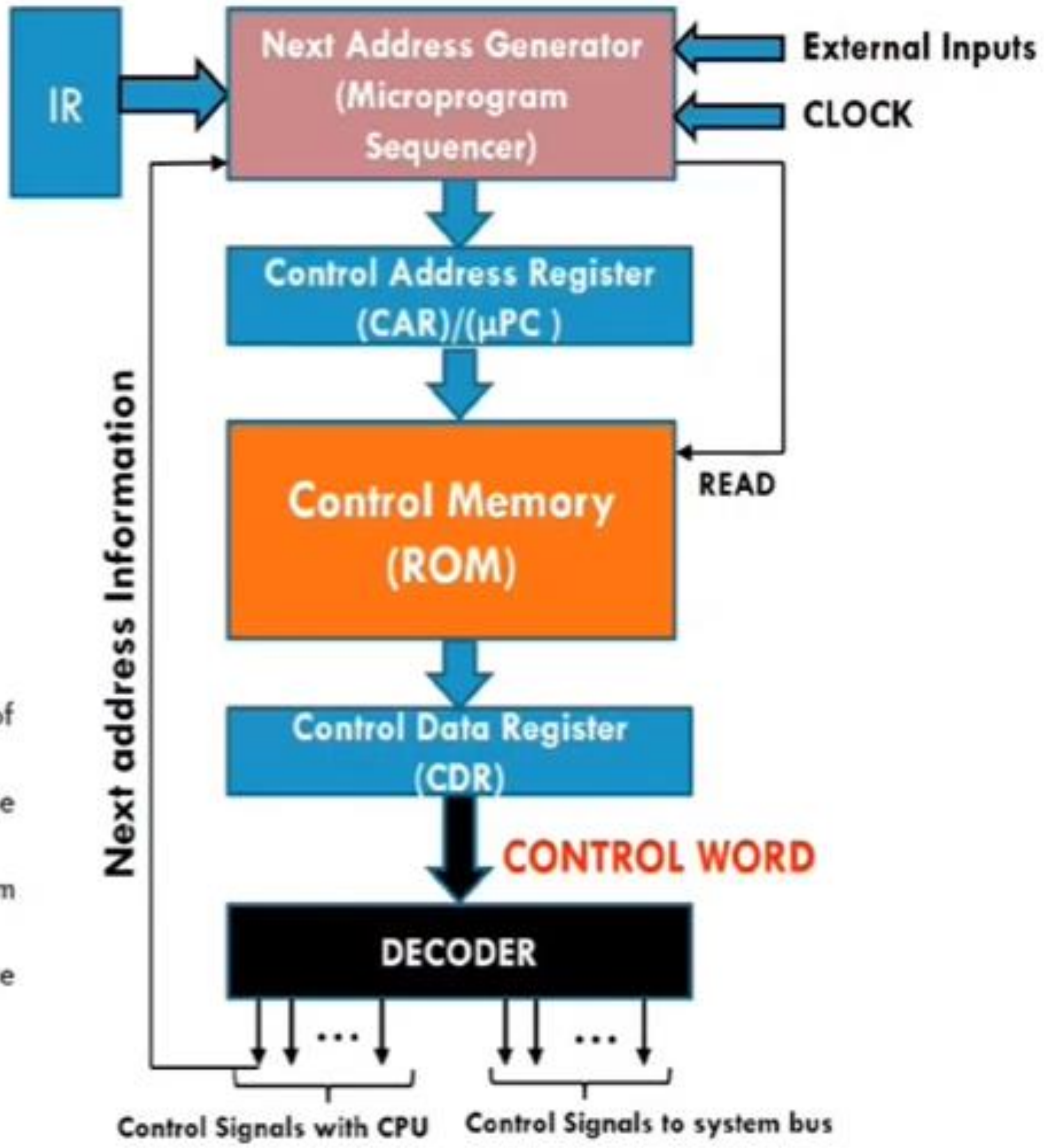
Control Memory:



- Tasks of Microprogrammed Control Unit:
  - Microinstruction sequencing
  - Microinstruction execution

The **Microprogram Sequencer** generates the address for microinstruction according to the instruction stored in the IR.

Microprogrammed Control Unit Block Diagram



Microprogrammed control unit consists of:

- **Control address register** – contains the address of the next microinstruction to be executed.
- **Control data register** – contains the microinstruction to be executed.
- **The sequencer** – determines the next address from within control memory
- **Control memory** – where microinstructions are stored.

Working

- The **next address generator** is called **Microprogram Sequencer** as it determines the address sequence of microinstructions that is read from control memory.
- The **Control Address Register (CAR)** holds address of next microinstruction to be read i.e. **µPC** (micro program counter).
- The **Control Data Register (CDR)** holds the microinstruction read from the control memory.
- **The Control Memory** contains set of microinstructions.
- **Microinstruction** contains a **control word** that specifies **one or more micro-operations** through generated **control signals.**
- When **address** is available in **Control Address Register** (CAR), the **sequencer** issues **READ** command to the **Control Memory**.
- After issue of READ command, the **word** from the addressed location is read into the **Control Data Register** (CDR).
- Now the **content of Control Data Register** (CDR) (i.e. microinstruction) generates ***control signal*** and ***next address information.***
- The **sequencer** loads a new address into the **Control Address Register (CAR)** based on the next address

Advantage

- The design of micro programmed control unit is **Simple (less complex)** because microprograms are implemented using **software routines.**
- Thus it is both **cheaper** and **less error-prone** to implement.
- **More flexible** because design, modification, correction and enhancement is easily possible.
- The **new or modified instruction set of CPU** can be **easily implemented** by simply rewriting or modifying the contents of control memory.
- **Complex function** such as floating-point arithmetic can be realized efficiently.
- **CISC (Complex Instruction Set Computer)** is implemented using Microprogrammed control unit.

Disadvantage

- The **micro programmed control** is **Slower** than **Hardwired control unit** because of time required to access the microinstructions from control memory.
- The flexibility is achieved at some **extra hardware cost due to control memory** and its access circuitry.

THANK YOU